

The screenshot shows a Pascal IDE with a project named 'labotarium 6'. The file explorer on the left lists various files, including 'sugar1.pas' through 'sugar10.pas'. The main editor displays the code for 'sugar1.pas', which is a program to calculate the total price of goods with a discount. The code is as follows:

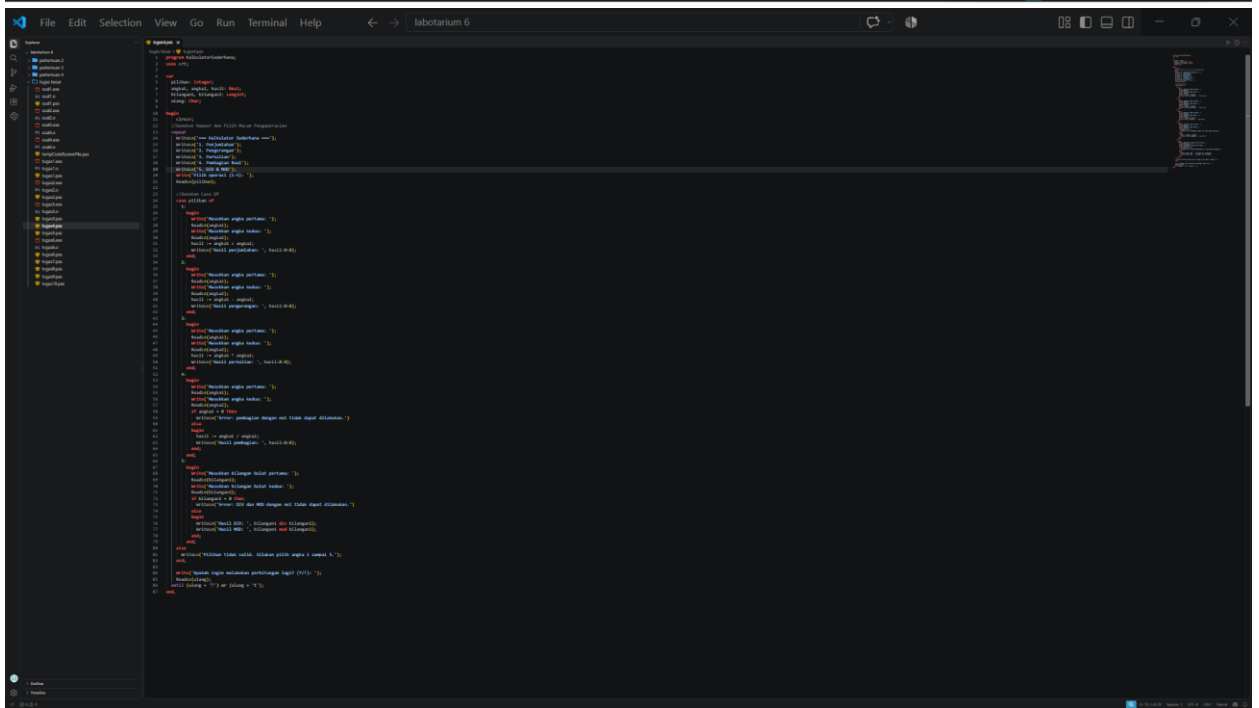
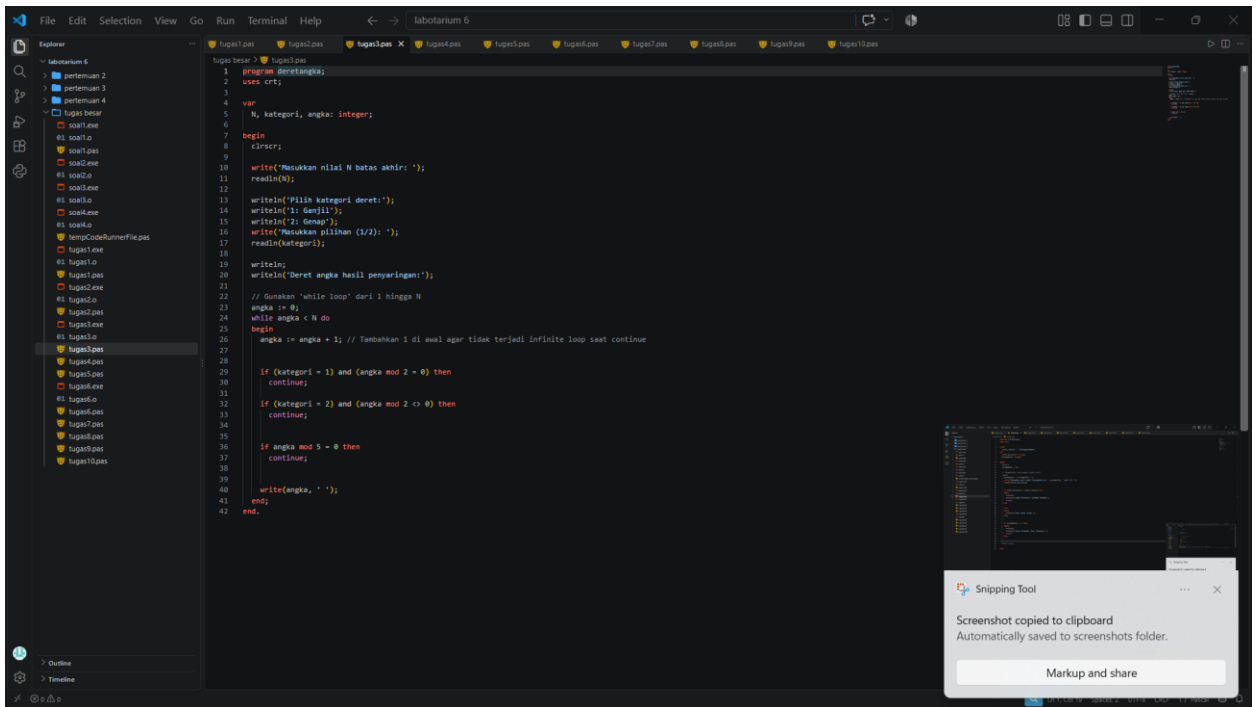
```
1 program labotarium6;  
2 uses crt;  
3  
4 var  
5   i, N, harga, total : longint;  
6   diskon, hargaAkhir : real;  
7  
8 begin  
9   clrscr;  
10  
11   writeln('Masukkan Jumlah Barang: ');  
12   readln(N);  
13  
14   for i := 1 to N do  
15   begin  
16     writeln('Masukkan Harga Barang ke-', i, ': ');  
17     readln(harga);  
18     total := total + harga;  
19   end;  
20  
21   writeln(total);  
22  
23   if total < 100000 then  
24     diskon := 0.0  
25   else if (total >= 100000) and (total < 500000) then  
26     diskon := 0.1  
27   else  
28     diskon := 0.2;  
29  
30   hargaAkhir := total - (total * diskon);  
31  
32   writeln('Jumlah Barang: ', N, ', Total: ', total, ', Diskon: ', diskon * 100, '%, Harga Akhir: ', hargaAkhir);  
33   writeln('Terima kasih ...');  
34 end.
```

1.

The screenshot shows a Pascal IDE with a project named 'labotarium 6'. The file explorer on the left lists various files, including 'sugar1.pas' through 'sugar10.pas'. The main editor displays the code for 'sugar1.pas', which is a program for a login system. The code is as follows:

```
1 program sistemlogin;  
2 uses crt;  
3  
4 const  
5   sandi_rahasia = 'utugasmumpuk';  
6  
7 var  
8   input_password : string;  
9   kesempatan : integer;  
10  
11 begin  
12   clrscr;  
13   kesempatan := 0;  
14  
15   // Menggunakan perulangan repeat-until  
16   repeat  
17     kesempatan := kesempatan + 1;  
18     writeln('Masukkan kata sandi (kesempatan ke-', kesempatan, ' dari 3): ');  
19     readln(input_password);  
20  
21     if input_password = sandi_rahasia then  
22     begin  
23       writeln;  
24       writeln('Login Berhasil! Selamat Datang!');  
25       break;  
26     end  
27   else  
28     writeln('Kata sandi salah.');
```

2.

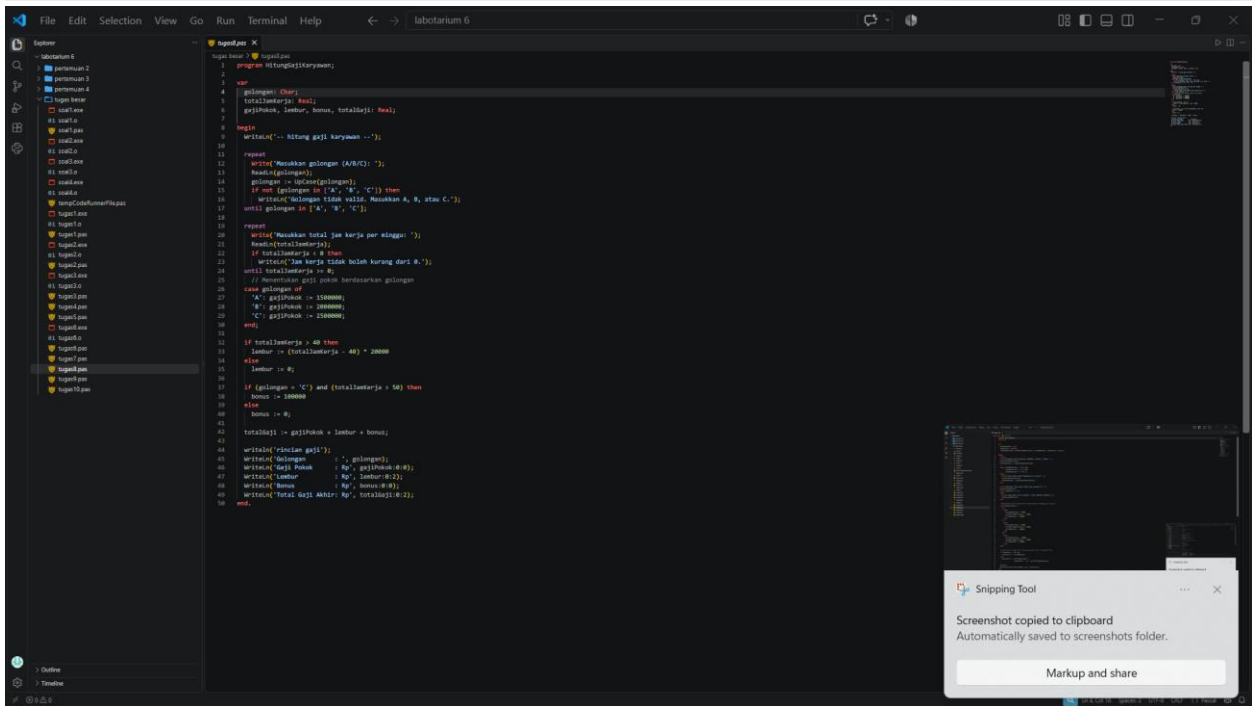
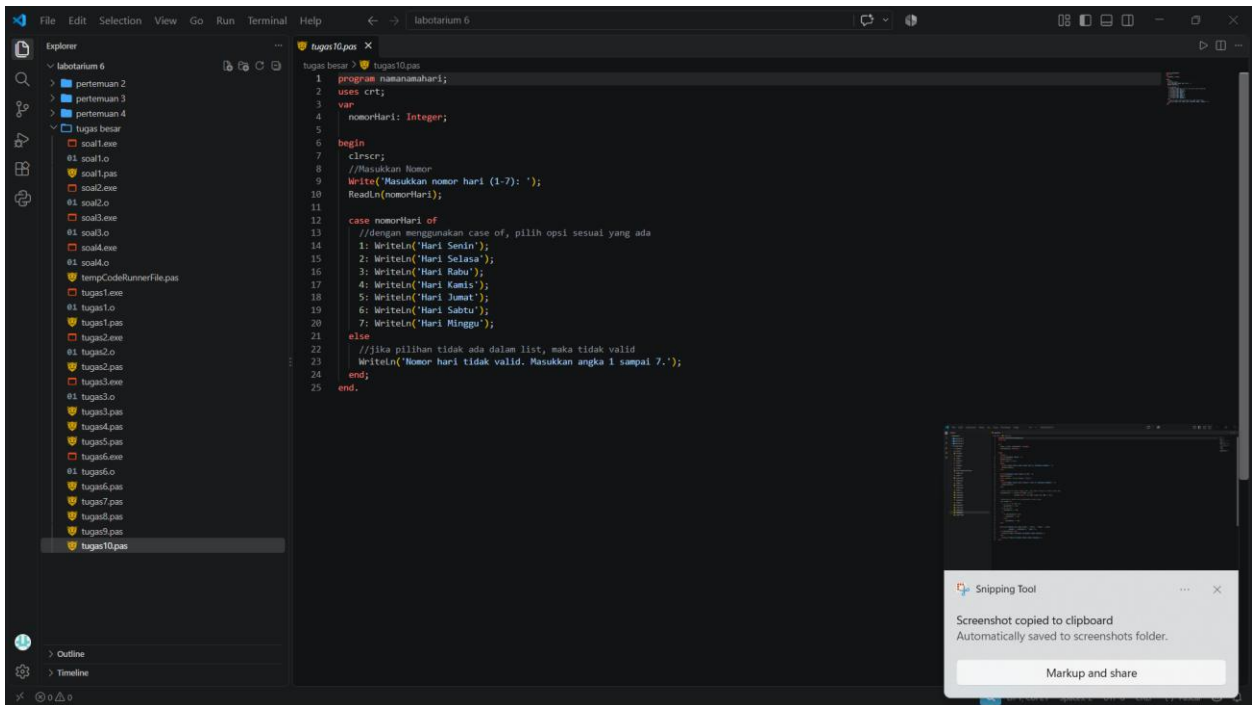


```
1 program PenilaianMetode;
2 uses crt;
3 var
4   tugas, uts, uas, kehadiran, nilai_akhir: real;
5   indeks: char;
6   status: string;
7
8 begin
9   clrscr;
10  writeln('--- program penilaian metode ---');
11
12  write('Masukkan Nilai Tugas (0-100) : ');
13  readln(tugas);
14  write('Masukkan Nilai UTS (0-100) : ');
15  readln(uts);
16  write('Masukkan Nilai UAS (0-100) : ');
17  readln(uas);
18  write('Masukkan Persentase Kehadiran (N) : ');
19  readln(kehadiran);
20
21  // Hitung Nilai Akhir (Tugas 30%, UTS 30%, UAS 40%)
22  nilai_akhir := (0.30 * tugas) + (0.30 * uts) + (0.40 * uas);
23
24  if nilai_akhir >= 85 then
25  begin
26    indeks := 'A';
27  end
28  else if nilai_akhir >= 75 then
29  begin
30    indeks := 'B';
31  end
32  else if nilai_akhir >= 60 then
33  begin
34    indeks := 'C';
35  end
36  else if nilai_akhir >= 50 then
37  begin
38    indeks := 'D';
39  end
40  else
41  begin
42    indeks := 'E';
43  end
44
45  if (nilai_akhir >= 60) and (kehadiran >= 80) then
46  begin
47    status := 'LULUS';
48  end
49  else
50  begin
51    status := 'TIDAK LULUS';
52  end
53
54  writeln('--- hasil penilaian ---');
55  writeln('Nilai Tugas : ', tugas:0:0);
56  writeln('Nilai Akhir : ', nilai_akhir:0:0);
57  writeln('Indeks Huruf : ', indeks);
58  writeln('Kehadiran : ', kehadiran:0:0, '%');
59  writeln('Status Kelulusan : ', status);
60 end;
```

5.

```
1 tugas_besar > huga6.pas
2 program penilaianmetode;
3 uses crt;
4 var
5   tugas, uts, uas, kehadiran, nilai_akhir: real;
6   indeks: char;
7   status: string;
8
9 begin
10  clrscr;
11  writeln('--- program penilaian metode ---');
12
13  write('Masukkan Nilai Tugas (0-100) : ');
14  readln(tugas);
15  write('Masukkan Nilai UTS (0-100) : ');
16  readln(uts);
17  write('Masukkan Nilai UAS (0-100) : ');
18  readln(uas);
19  write('Masukkan Persentase Kehadiran (N) : ');
20  readln(kehadiran);
21
22  // Hitung Nilai Akhir (Tugas 30%, UTS 30%, UAS 40%)
23  nilai_akhir := (0.30 * tugas) + (0.30 * uts) + (0.40 * uas);
24
25  if nilai_akhir >= 85 then
26  begin
27    indeks := 'A';
28  end
29  else if nilai_akhir >= 75 then
30  begin
31    indeks := 'B';
32  end
33  else if nilai_akhir >= 60 then
34  begin
35    indeks := 'C';
36  end
37  else if nilai_akhir >= 50 then
38  begin
39    indeks := 'D';
40  end
41  else
42  begin
43    indeks := 'E';
44  end
45
46  if (nilai_akhir >= 60) and (kehadiran >= 80) then
47  begin
48    status := 'LULUS';
49  end
50  else
51  begin
52    status := 'TIDAK LULUS';
53  end
54
55  writeln('--- hasil penilaian ---');
56  writeln('Nilai Tugas : ', tugas:0:0);
57  writeln('Nilai Akhir : ', nilai_akhir:0:0);
58  writeln('Indeks Huruf : ', indeks);
59  writeln('Kehadiran : ', kehadiran:0:0, '%');
60  writeln('Status Kelulusan : ', status);
61 end;
```

6.



```
1 program jumlahhariindonesia;
2 uses crt;
3
4 var
5     tahun, bulan, jumlahHari: Integer;
6     tahunKabisat: Boolean;
7
8 begin
9     clrscr;
10    write('Masukkan tahun: ');
11    ReadLn(tahun);
12    while tahun < 0 do
13    begin
14        write('Tahun harus lebih besar dari 0. Masukkan kembali: ');
15        ReadLn(tahun);
16    end;
17
18    write('Masukkan nomor bulan (1-12): ');
19    ReadLn(bulan);
20    while (bulan < 1) or (bulan > 12) do
21    begin
22        write('Nomor bulan harus antara 1 dan 12. Masukkan kembali: ');
23        ReadLn(bulan);
24    end;
25
26    //Tahun kabisat habis dibagi 400, atau habis dibagi 4 tetapi tidak 100
27    tahunKabisat := (tahun mod 400 = 0) or
28                  ((tahun mod 4 = 0) and (tahun mod 100 <> 0));
29
30    //Menentukan jumlah hari berdasarkan nomor bulan
31    case bulan of
32        1, 3, 5, 7, 8, 10, 12:
33            jumlahHari := 31;
34        4, 6, 9, 11:
35            jumlahHari := 30;
36        2:
37            if tahunKabisat then
38                jumlahHari := 29
39            else
40                jumlahHari := 28;
41    end;
42
43    write('Berikut hari pada bulan ', bulan, ' tahun ', tahun,
44          ' adalah: ', jumlahHari, ' hari.');
```

9.

```
1 program namamamahari;
2 uses crt;
3
4 var
5     nomorHari: Integer;
6
7 begin
8     clrscr;
9     //Masukkan Nomor
10    Write('Masukkan nomor hari (1-7): ');
11    ReadLn(nomorHari);
12
13    case nomorHari of
14        //dengan menggunakan case of, pilih opsi sesuai yang ada
15        1: WriteLn('Hari Senin');
16        2: WriteLn('Hari Selasa');
17        3: WriteLn('Hari Rabu');
18        4: WriteLn('Hari Kamis');
19        5: WriteLn('Hari Jumat');
20        6: WriteLn('Hari Sabtu');
21        7: WriteLn('Hari Minggu');
22    else
23        //jika pilihan tidak ada dalam list, maka tidak valid
24        WriteLn('Nomor hari tidak valid. Masukkan angka 1 sampai 7.');
```

10.